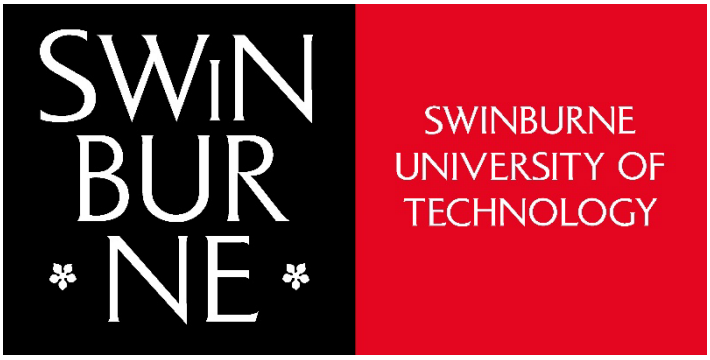


Unit Code: COS30049

Unit Name: Computing Technology Innovation Project

Project Proposal: Smart Ground-Truthing and Digital Biodiversity System for Plant Species Documentation

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TABLE OF CONTENTS

Table of Contents i
List of Figures ii
List of Tables ii
1 Background 1
1.1 Problem Statement 2
1.2 Existing Solutions 3
1.3 Motivations 4
2 Objectives and Scope 5
2.1 Project Objectives 5
2.2 Scope 5
2.3 Secure Software Development Lifecycle 6
2.4 Risk Register 7
2.5 Constraints 9
3 Stakeholders 9
3.1 System Roles and Interactions 10
3.2 Feasibility Evaluation 11
4 Deliverables 12
4.1 Initial Release Schedule of the Product Backlog Items 12
4.2 Diagrams 14
4.2.1 Hardware Diagram 14
4.2.2 Software Diagram 15
4.2.3 Activity Diagram 17
4.2.4 Use Cases Diagram 18
4.3 Alternative Solutions 19
5 Quality Management 21
5.1 Priority Quality Characteristics 21
5.2 Quality Assurance Process 22
5.3 Quality Attributes, SMART Measurement and Definition of Done 23
6 Resources 27
6.1 Human Resources 27
6.1.1 Knowledge 27

6.1.2 Abilities and Skills 27
6.1.3 Attitudes 27
6.1.4 Energy 27
6.1.5 Time 27
6.2 Software Resources 28
6.3 Hardware Resources 29
6.4 Cost Analysis of Prototypes 30
7 Approval Signatures 32
Project Team 32
Project Sponsor [Ts. Dr. Lee Sue Han] 33
8 References 34

LIST OF FIGURES

Figure 1: Hardware Architecture Diagram 14
Figure 2: Software Architecture Diagram 15
Figure 3: Activity Diagram 17
Figure 4: Use Case Diagram 18

LIST OF TABLES

Table 1: Review of existing solutions 3
Table 2: Project objectives 5
Table 3: Secure Software Development Lifecycle activities and artefacts 7
Table 4: Risk register 9
Table 5: Project constraints 9
Table 6: System roles and their interactions 11
Table 7: Schedule of Deliverables 12
Table 8: Initial release schedule of the product backlog items 13
Table 9: Alternative Solutions 20
Table 10: Quality attributes, SMART measurement and Definition of Done for each backlog item 26
Table 11: Software Resources 29
Table 12: Hardware Resources 30
Table 13: Cost Analysis of Prototypes 31

1 BACKGROUND

Biodiversity conservation plays a crucial role in preserving ecosystem stability, supporting scientific research, and promoting sustainable environmental management (Cardinale et al., 2012). Accurate plant species documentation is essential for monitoring biodiversity, assessing conservation status, conducting ecological studies, and facilitating environmental education (Heberling et al., 2021). In protected forest ecosystems such as those in Sarawak, comprehensive biodiversity records provide valuable information for conservation planning, habitat management, and ecotourism development.

The advancement of digital technologies, including Artificial Intelligence (AI), Internet of Things (IoT), cloud computing, and mobile applications, has created new opportunities to improve biodiversity management practices (Allan et al., 2018). As a Sarawak-based technology company specialising in AI, computer vision, machine learning, and IoT solutions, NeuonAI Sdn Bhd focuses on developing innovative technologies that transform research outcomes into practical industrial and environmental applications. Established in December 2018, the company provides AI-driven solutions for industries such as manufacturing, agriculture, and system management, and its team won the top three places in the PlantCLEF 2019 plant identification challenge, with a strong emphasis on intelligent data management and digital transformation (Dayangku, 2020). Through its collaboration with Sarawak Forestry Corporation (SFC) as a commercialisation partner, NeuonAI contributes to the development of digital biodiversity management initiatives that support conservation and environmental sustainability (NEUON AI, n.d.; The Edge Malaysia, 2026).

Despite the increasing availability of digital technologies, plant species documentation within many conservation areas continues to rely heavily on conventional methods. Field personnel are often required to identify plant species, attach physical labels, record observations manually, capture photographs separately, and subsequently transfer collected information into digital databases. These processes are labour-intensive, time-consuming, and susceptible to data inconsistencies, duplication, and information loss (Barchard & Pace, 2011; Thriemer et al., 2012). Furthermore, biodiversity surveys are frequently conducted in remote forest environments where internet connectivity is limited, making real-time data collection and synchronisation challenging (Wong, 2026).

In addition, biodiversity information is often dispersed across multiple records and repositories, limiting accessibility for researchers, conservation officers, local communities, and visitors. The lack of a centralised and searchable biodiversity information system hinders knowledge sharing, reduces operational efficiency, and restricts opportunities for biodiversity education and ecotourism engagement. Digital biodiversity platforms have been identified as important tools for improving data accessibility, supporting informed conservation decisions, and facilitating public participation in biodiversity conservation (Heberling et al., 2021).

To address these challenges, this project proposes the development of a Smart Ground-Truthing and Digital Biodiversity System for Plant Species Documentation. The proposed system integrates a mobile-based biodiversity documentation platform with QR code

technology, GPS-assisted location tracking, offline data collection capabilities, and cloud synchronisation. The system is complemented by a web-based digital plant knowledge repository that provides access to plant taxonomy, species descriptions, conservation status, distribution information, and supporting multimedia content. Additionally, cybersecurity measures are incorporated to ensure secure biodiversity data management, while IoT-enabled monitoring technologies are employed to support the protection of endangered plant species through real-time environmental monitoring and threat detection.

By providing an integrated digital ecosystem for plant species documentation and monitoring, the proposed system aims to improve the efficiency and accuracy of biodiversity data collection, enhance accessibility to biodiversity information, strengthen conservation management practices, and support sustainable biodiversity conservation initiatives within Sarawak's protected forest environments.

1.1 Problem Statement

Plant documentation in Sarawak's protected areas still relies mainly on handwritten records, physical tags and repeated data entry. The process is slow and labour-intensive, and it is prone to human error (Barchard & Pace, 2011). The result is duplicate records, uneven data quality and a real risk of losing information. As the number of recorded plants grows, the data becomes harder to manage and less useful for monitoring.

In addition, access to biodiversity information remains limited among key stakeholders, including researchers, conservation officers, local communities, and visitors. Biodiversity records are often dispersed across multiple sources, making it difficult to obtain accurate and up-to-date information when required. The lack of a centralised and accessible digital platform restricts knowledge sharing, reduces the efficiency of conservation management activities, and limits opportunities for scientific research, environmental education, and biodiversity-based ecotourism development.

1.2 Existing Solutions

Several existing tools already solve parts of this problem. Each was reviewed to establish whether an existing product could satisfy the requirements before deciding to build a custom system.

Existing Solution	Capability	Limitation for This Project
iNaturalist / Pl@ntNet	Citizen-science platforms offering photograph-based species suggestion, observation logging and community verification.	Observation-centric rather than specimen-centric: they record sightings, not a persistent identity for an individual tagged plant that is revisited over years. Data is held on third-party infrastructure outside SFC control, and although Pl@ntNet now offers an offline identification mode (Pl@ntNet, 2022), neither provides an agency-controlled offline capture, review and synchronisation workflow.
GBIF	Global aggregator of biodiversity occurrence records with standardised Darwin Core exchange formats, supporting large-scale biodiversity synthesis (Heberling et al., 2021).	A publication and aggregation layer, not a field collection tool. It consumes cleaned datasets rather than producing them, and offers no tagging, offline capture or monitoring capability.
Open Data Kit, KoBoToolbox, Epicollect5	Configurable offline-first mobile survey platforms with form design, GPS capture, photograph attachment and deferred synchronisation (Hartung et al., 2010).	Generic form builders. They support data capture well but provide no QR-anchored specimen identity, no public-facing knowledge repository, no role-based review and approval workflow, and no IoT integration. Custom dashboards and reporting are constrained by the platform.
Symbiota	Open-source platform for herbarium and biodiversity collection portals with taxonomic management and distribution mapping (Gries et al., 2014).	Designed around herbarium specimens and institutional collections rather than live in-situ plants under active field monitoring. No mobile offline capture and no sensor integration.
Generic QR asset-tagging tools	Commercial asset management products that bind QR codes to database records.	Built for inventory and maintenance domains. No taxonomic data model, no conservation status handling, and no ecological or environmental monitoring features.

Table 1: Review of existing solutions

None of these products covers all four things this project needs: offline field capture linked to a permanent QR identity for each plant; a plant knowledge repository that SFC controls, with a review and approval step; security suited to the locations of protected species; and IoT

monitoring of endangered plants. Each tool handles one or two of these at most, which justifies a custom system. The review also shows that we should follow existing standards such as Darwin Core (Wieczorek et al., 2012), so that SFC can later publish its data to platforms like GBIF.

1.3 Motivations

- Operational efficiency: recording data once, directly on a phone, removes the need to copy paper forms into a database. This saves botanists time and cuts transcription errors (Barchard & Pace, 2011; Thriemer et al., 2012).
- Data integrity and continuity: because each plant keeps the same QR identity, different officers can revisit and update the same individual across many surveys.
- Conservation outcomes: sensors around rare plants can flag disturbance as it happens, so staff can respond sooner to threats such as poaching or habitat damage. These threats tend to get less attention for plants than for animals (Margulies et al., 2019).
- Knowledge accessibility: one searchable repository serves researchers, park guides and visitors, supporting both research and ecotourism.
- Academic and professional development: the project covers mobile and web development, backend engineering, embedded IoT, applied machine learning and cybersecurity, giving each of us hands-on experience with a complete system.

2 OBJECTIVES AND SCOPE

2.1 Project Objectives

The project is directed by five objectives, the first four of which are required by the brief and the fifth is optional.

No.	Objective	Description
1	Develop a Smart Digital Ground-Truthing Platform	Deliver an integrated mobile and web biodiversity documentation platform supporting QR code scanning and binding, plant species recording, offline field data collection, GPS-based location capture, and synchronisation with a centralised biodiversity database.
2	Develop a Digital Plant Knowledge System	Establish a searchable digital repository containing plant taxonomy, ecological information, conservation status, botanical photographs, species distribution maps and cultural significance, supporting researchers, conservation officers, park guides and visitors.
3	Ensure Secure Biodiversity Data Management	Implement secure authentication, encrypted data storage and transmission, role-based access control and secure cloud synchronisation to protect biodiversity records, protected-species location data and user information.
4	Implement IoT-Based Plant Protection	Develop an IoT-enabled monitoring prototype that collects real-time environmental and location data, detects potential threats such as physical disturbance or unauthorised presence, and supports rapid conservation response for protected plant species.
5	Develop an Innovative Ground-Truthing Workflow (Optional)	Design and evaluate an improved approach to plant ground-truthing that increases the efficiency, accuracy or scalability of field data collection in remote forest environments with limited connectivity.

Table 2: Project objectives

2.2 Scope

This project aims to replace the paper-based process for documenting plants in Niah National Park with a digital one, working with SFC and NEUON AI. Botanists and conservation officers will use a mobile app to record plants, capture GPS locations and photos, and scan or generate a QR code for each plant. The app works offline, because connectivity is poor in forests like Niah (Wong, 2026), which covers around 3,140 hectares (Meekiong & Wasli, 2025).

A Digital Plant Knowledge System website will let conservation officers add, update, view and export plant data such as taxonomy, conservation status, images and location. Researchers, guides and visitors will be able to search and browse the same information. Security features, including authentication, access control and encryption, will be built in by following a Secure Software Development Life Cycle (SSDLC), with rounds of vulnerability assessment and fixes (Souppaya et al., 2022). An IoT prototype will also monitor the location and environmental conditions around selected plants.

Some things are out of scope. The system will be piloted in Niah National Park only and will not be rolled out across SFC's wider network of protected areas (The Borneo Post, 2026). It will not identify species automatically from photos, because image-based plant recognition is part of NEUON AI's own R&D (Dayangku, 2020) and SFC is developing a separate image-recognition app (The Borneo Post, 2026). Botanists will enter the species themselves. The IoT component will be built and tested as a prototype only, without buying, calibrating or deploying sensors in the park. The project will also not connect to SFC's or other organisations' existing databases, including the UNESCO documentation linked to Niah's World Heritage listing (The Star, 2024; UNESCO World Heritage Centre, n.d.).

2.3 Secure Software Development Lifecycle

Because the system holds personal user data and the precise coordinates of protected species, security is integrated into every phase of development rather than tested at the end. The team follows a Secure Software Development Lifecycle, with Nicole Jia Xin TAN owning security architecture and Michelle Mei Xuan TAN owning security testing.

SSDLC Phase	**Activity**	**Artefact Produced**
Requirements	Define security requirements alongside functional requirements: role definitions and permission matrix, data classification, encryption requirements, and coordinate-precision policy for public views.	Security requirements register and data classification table (Sprint 1, week 1).
Design	Threat modelling of the attack surface using STRIDE, covering the mobile-to-API boundary, the MQTT broker, the synchronisation endpoint and the public visitor interface.	Threat model with identified threats and planned countermeasures (Sprint 1).
Development	Apply secure coding practices: input validation, parameterised queries, output encoding, HTTPS and TLS enforcement, secrets kept out of version control, and dependency vulnerability scanning.	Secure coding checklist applied at code review; dependency scan reports (both sprints).
Verification	Conduct the application vulnerability assessment across all endpoints and interfaces using OWASP ZAP for dynamic analysis, npm audit for dependency analysis and manual authorisation testing, scoped to the web application, the REST API, the authentication flow and the MQTT broker configuration,	Vulnerability assessment report with findings rated by severity (Sprint 2, week 1).

	with findings classified against the OWASP Top 10 (Open Web Application Security Project, 2021).	
Remediation	Prioritise findings by severity and remediate, addressing authentication and authorisation defects, input validation gaps, access control issues and encryption weaknesses.	Remediation log mapping each finding to its fix and the commit that implemented it (Sprint 2, week 2).
Reassessment	Re-run the full assessment using the same tools and scope to verify that identified vulnerabilities have been closed and that no regressions were introduced.	Reassessment report evidencing resolution, with before-and-after comparison (Sprint 2, week 3).

Table 3: Secure Software Development Lifecycle activities and artefacts

Scheduling the assessment at the start of Sprint 2 rather than at the end is deliberate: the brief requires evidence of reassessment after remediation, which is only possible if the assess, remediate and reassess cycle is given room to complete before final submission.

2.4 Risk Register

Risks are rated for likelihood and impact on a three-point scale. Each risk is assigned an owner responsible for monitoring it and triggering the mitigation.

ID	Risk	Likelihood / Impact	Mitigation	Owner
R1	React learning curve delays Sprint 1 delivery	High / High	Dedicate the first three days of Sprint 1 to a shared upskilling task with a small reference implementation. Pair the three members with Vue experience alongside those without. Keep Sprint 1 scope to core flows only.	Scrum Master
R2	Offline synchronisation produces data loss or duplicate records	Medium / High	Assign to the member with prior local-database experience. Use server-assigned unique identifiers with client-side idempotency keys. Write synchronisation test cases before implementation and include a dedicated offline test scenario in the test strategy.	Albian (PO)
R3	ESP32 sensor node unreliable or sensors faulty	Medium / Medium	Order spare sensors early. Build a software sensor simulator that publishes synthetic telemetry to the broker so that backend, dashboard and alerting work can	Amberlyn

			proceed independently of hardware availability.	
R4	Supabase free-tier limits constrain storage or bandwidth	Low / Medium	Compress and resize photographs on the device before upload. Monitor usage against tier limits. Retain a local PostgreSQL instance as a fallback for demonstration.	Patrick
R5	Vulnerability assessment uncovers defects too late to remediate	Medium / High	Schedule the assessment at the start of Sprint 2 rather than the end. Run dependency scanning continuously from Sprint 1 so that routine issues are cleared before the formal assessment.	Michelle
R6	Scope creep from the optional innovation component	Medium / Medium	Place all optional items last in the backlog with an explicit descope trigger: if any mandatory item is incomplete entering the final two weeks, optional items are dropped without further discussion.	Product Owner
R7	Team member unavailable through illness or competing assessment load	Medium / Medium	Maintain a documented secondary owner for each workstream. Keep design decisions and setup steps in the shared repository rather than in individual knowledge.	Scrum Master
R8	Integration between independently built components fails late	Medium / High	Agree the API contract in Sprint 1 week 1 and publish it in the repository. Build a deliberately thin end-to-end vertical slice early rather than completing components in isolation.	Patrick
R9	GPS accuracy under forest canopy insufficient to distinguish adjacent specimens	High / Low	Specimen identity is anchored to the QR code rather than to coordinates, so accuracy affects mapping quality only. Record the device-reported accuracy value with each fix and display it, so that map precision is represented honestly.	Ricardo
R10	Anomaly detection produces excessive false alerts	Medium / Medium	Collect a baseline period of readings before fixing thresholds. Make thresholds configurable per node from the administrator dashboard rather than hard-coded. Require sustained rather than instantaneous threshold breach before raising an alert.	Norman

R11	Protected-species coordinates exposed through the public visitor interface	Low / High	Coordinate precision is deliberately reduced for any record flagged as endangered before it reaches a public view. Enforce this at the API layer rather than in the client, and include it as a test case in the security assessment.	Nicole

Table 4: Risk register

2.5 Constraints

Constraint	Description and Implication
Limited connectivity at the pilot site	Niah National Park has unreliable mobile and wireless coverage. All field-facing functionality must operate fully offline and synchronise opportunistically. This constraint drives the local storage design, the token-based authentication choice and the selection of MQTT for telemetry.
Project timeline	Approximately seven development weeks are available after proposal submission, delivered across two sprints. This caps the achievable scope and requires optional items to be descopable without affecting mandatory objectives.
Team skill profile	No member has prior React, Supabase or MQTT experience. Upskilling time is budgeted explicitly in Sprint 1 rather than assumed to be absorbed.
Budget	No external funding is available. Hardware is limited to low-cost components and equipment already held by the team or available in the IoT laboratory, and all software must be open source or free-tier.
Hardware availability	The IoT node is a bench and short-range field prototype. Weatherproofing, solar power and long-range backhaul are not affordable within the budget and are excluded.
Data sensitivity	The system holds personal data and protected-species coordinates. Security requirements constrain the design of every interface, particularly the public-facing visitor view.
Absence of a labelled training dataset	No labelled dataset exists for either Sarawak plant species imagery or genuine disturbance events. This constrains the anomaly detection approach to threshold and statistical methods and excludes species classification entirely.
Device platform	Testing is conducted on Android devices available to the team. iOS behaviour is not verified.

Table 5: Project constraints

3 STAKEHOLDERS

The main stakeholders in the project are the Sarawak Forestry Corporation (SFC), the owner of the project and the data that is used in the project where the company manages 71 national parks in the state of Sarawak (The Borneo Post, 2026), and botanists and conservation officers who will utilize the system out in the field. These field officers are the primary users

of the application on their smartphones. They will use the mobile application to log information about plants, GPS locations, and synchronize information that they have obtained during field work in Niah National Park.

There are also other stakeholders who are interested in this project. One of them is NEUON AI, which is an industry partner, having AI and plant recognition knowledge, and collaborating with SFC (Dayangku, 2020; The Edge Malaysia, 2026). In addition, researchers will get easy access to plant data (Heberling et al., 2021). Moreover, both park guides and visitors will be able to get easy access to educational information about plants. Dr Sue Lee Han, being a tutor of the project, will verify academic quality of the project and approve milestones. It is the responsibility of us, the student team to develop, test and document the system. The system administrators will ensure system security after the handover of the system to them.

3.1 System Roles and Interactions

Four roles will interact with the system. Three are authenticated and the fourth is an unauthenticated public role. We kept the Conservation Officer and System Administrator as separate roles because the brief treats managing plant content and running the platform as different jobs.

Role	**Interactions with the System**
Botanist	Registers an account and authenticates on the mobile application. Scans an unassigned QR tag and binds it to a new specimen record or generates a new QR code where no pre-printed tag is available. Records taxonomy, morphological characteristics, height and related attributes. Captures and attaches photographs and records GPS coordinates automatically. Saves records to local device storage while offline and synchronises them when connectivity returns. Reviews personal submission history and the approval status of past records.
Conservation Officer	Adds, edits and deletes plant species records in the knowledge system. Uploads botanical photographs and supporting taxonomic information. Reviews submitted field observations and approves, rejects or returns them for correction. Searches and filters biodiversity records by scientific name, common name, conservation status and location. Views species distribution maps generated from approved observation coordinates. Generates and exports biodiversity reports.
System Administrator	Creates, edits and deactivates user accounts and assigns roles. Registers IoT sensor nodes and associates them with monitored specimens. Monitors the real-time environmental dashboard and receives automated threat alerts. Acknowledges alerts, dispatches investigation and records case outcomes. Reviews audit logs and manages security configuration.

Visitor (unauthenticated)	Scans the public QR tag on a tagged specimen to retrieve its published plant profile. Browses and searches published species information in the public knowledge system. Cannot view unapproved submissions, precise coordinates of endangered specimens or any user data.

Table 6: System roles and their interactions

3.2 Feasibility Evaluation

1. **Feasibility from a technical perspective.**

It will be possible to implement the system via existing and familiar technologies. Offline storage with later sync, QR codes, GPS tracking and secure login are all common in mobile apps and carry little technical risk (Hartung et al., 2010). The main challenge is making the app work reliably offline, given the poor connectivity at Niah National Park (Wong, 2026). This aspect is familiar for mobile applications; therefore, it is manageable rather than the major risk. The IoT component of the project will be just a prototype rather than a functional field solution.

2. **Feasibility of operation.**

The system fits SFC's existing work, since SFC already carries out conservation and monitoring in its parks (The Borneo Post, 2026). The app supports what officers already do rather than changing their role. Covering a single park keeps rollout and maintenance simple, and because the officers who will use the system are also stakeholders, they have a direct say in how it works.

3. **Feasibility of legal and ethics.**

The system will hold users' personal data, field records and the locations of protected species, which could help poachers if leaked (Lindenmayer & Scheele, 2017). This is feasible as long as security is designed in from the start through secure login, encryption and access control, following an SSDLC approach (Souppaya et al., 2022). Users' personal data must also be handled in line with Malaysia's Personal Data Protection Act 2010, as amended in 2024 (Personal Data Protection Department, n.d.). We do not expect other legal or regulatory issues, because the system will not connect to SFC's production systems or to UNESCO's heritage documentation for Niah (The Star, 2024).

4 DELIVERABLES

This section lists what the team will deliver and when. The deliverables follow the development lifecycle, from planning and requirements, through implementation, to testing, documentation and handover. The schedule is based on the unit's teaching weeks and is ordered so that later work never waits on unfinished groundwork.

The deliverables listed below reflects the current project scope and may change as the project progresses and as feedback is received from the supervisor and stakeholders.

Deliverable	Description	Schedule
Project scope	A defined statement of project boundaries, inclusions, exclusions, stakeholders, and feasibility evaluation, forming the agreed baseline for subsequent work.	Week 3
Task delegation	An allocation of roles and responsibilities across team members, including ownership of the mobile application, web system, IoT prototype, and security workstreams.	Week 3
Basic code structure	An initial project skeleton covering repository setup, architectural layering, database schema drafts, and shared development conventions.	Week 4
All source code	The complete implementation of the mobile ground-truthing application, the Digital Plant Knowledge System, the synchronisation layer, and the IoT simulation module.	Week 11
Software testing	Documented functional, integration, usability, security, and performance testing evidence, including defect logs and remediation records.	Week 11
User manual	End-user documentation covering field data collection workflows, web system administration, and troubleshooting guidance for conservation officers and administrators.	Week 11

Table 7: Schedule of Deliverables

4.1 Initial Release Schedule of the Product Backlog Items

The backlog is ordered by technical dependency. Sprint 1 covers the shared architecture, user login and the core field app workflow, including plant recording and QR scanning. Sprint 2 builds on these with the web plant database and the IoT components. Table 8 lists each backlog item with its sprint and target week.

#	Product Backlog Item	Sprint	Target Week
1	Project foundation, requirements baseline and shared architecture	Sprint 1	Week 4
2	Authentication and role-based access control	Sprint 1	Week 7
3	Mobile plant observation with local SQLite storage	Sprint 1	Week 7
4	QR scanning, photographs and GPS capture	Sprint 1	Week 7
5	Cloud synchronisation and conflict/error handling	Sprint 2	Week 11
6	Web plant database, search and filtering	Sprint 2	Week 11
7	Observation review, mapping and reporting	Sprint 2	Week 11
8	ESP32 IoT sensing, MQTT ingestion and data storage	Sprint 2	Week 11

9	IoT dashboard, threat detection and automated alerts	Sprint 2	Week 11
10	Security assessment, usability testing and integrated demonstration	Sprint 2	Week 11
11	*(Optional)* Ground-truthing innovation and observation quality indicator	Sprint 2	Week 11

Table 8: Initial release schedule of the product backlog items

4.2 Diagrams

4.2.1 Hardware Diagram

The hardware diagram depicts the components of the Smart Ground-Truthing and Digital Biodiversity System and how each unit interacts with the overall system.

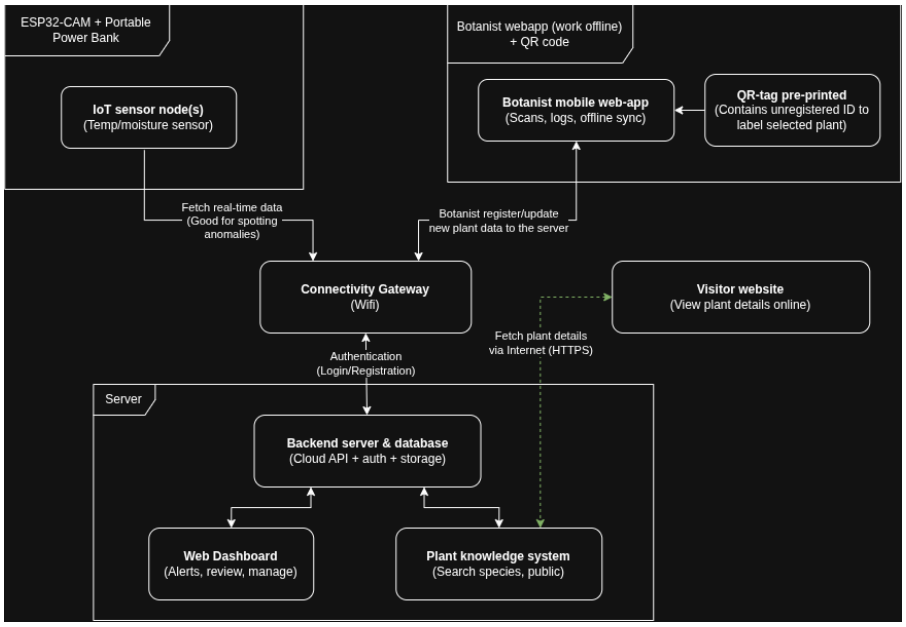


Figure 1: Hardware Architecture Diagram

Hardware.drawio

As shown in the hardware diagram above, the IoT sensor node(s) and Botanist mobile app (React Native) serve as the primary field components of the system. The IoT sensor node integrates an ESP32-CAM unit with temperature, humidity, soil moisture, motion (PIR) and GPS sensors, powered by a portable power bank for field deployment. It continuously collects real-time telemetry useful for spotting anomalies and transmits data upstream via a Wi-Fi Connectivity Gateway using the MQTT protocol. Complementing the sensors, pre-printed QR tags containing unregistered IDs are placed onto selected plants. Botanists use the Botanist mobile app (React Native) to scan tags, log or update plant data, and perform synchronization with the server when the web-app is back online. Park visitors access the Visitor website over HTTPS to view plant details online.

All central data processing, authentication, and storage are managed by the Backend server & database (cloud API, auth, and storage) through the Connectivity Gateway. The backend server receives real-time IoT sensor telemetry and updates from the Botanist application after authentication (login/registration). When data is processed, administrators monitor alerts, review, and manage records via the Web Dashboard. Verified species records are published to the public Plant knowledge system, allowing visitors on the Visitor website to search species and view botanical information.

4.2.2 Software Diagram

The software diagram illustrates the system architecture of the Smart Ground-Truthing and Digital Biodiversity System, divided into three distinct layers: the Frontend/UI Layer, the Network Layer, and the Backend.

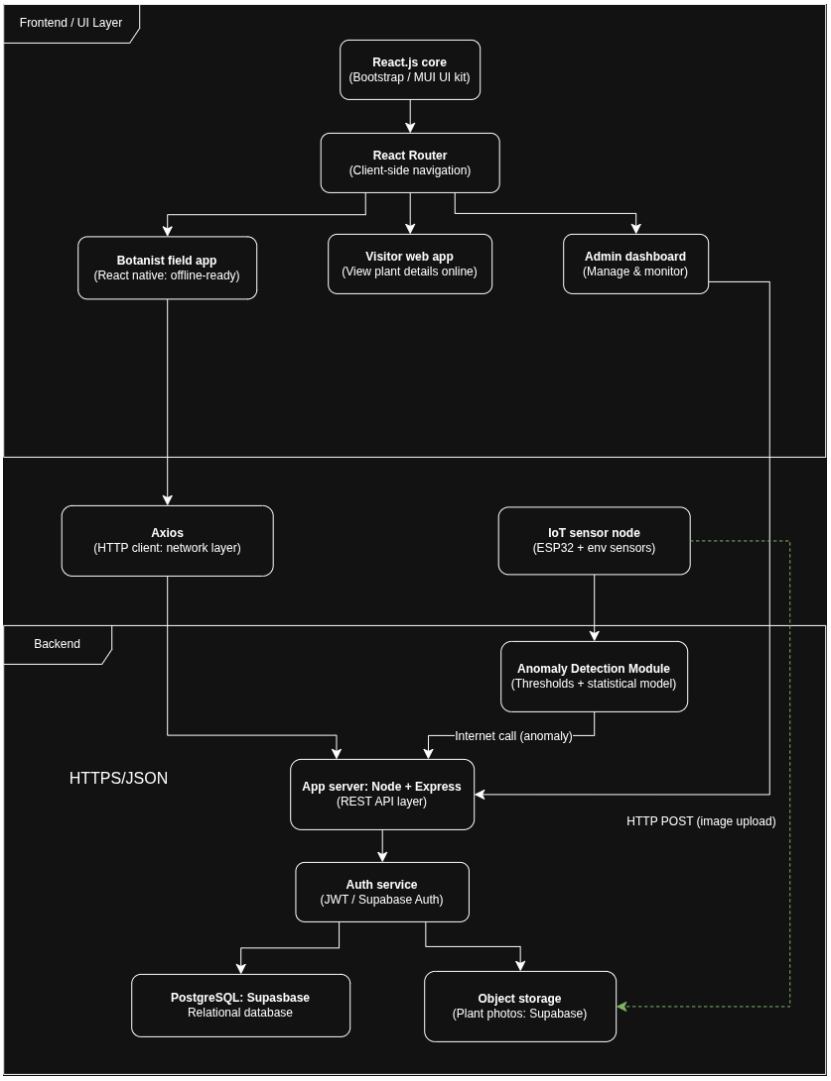


Figure 2: Software Architecture Diagram

Software.drawio

The Frontend/UI Layer utilizes React.js core supported by Bootstrap and MUI UI kits for clean, consistent interface styling, with React Router directing client-side navigation. This layer supports three dedicated client applications: the offline-ready Botanist Field App built on React Native for field data collection and QR scanning, the Visitor App for plant taxonomy and biodiversity information search, and the Admin Dashboard for central data management and monitoring.

To facilitate communication between the frontend clients and the server infrastructure, Axios acts as the HTTP client in the Network Layer, transmitting requests and handling responses via HTTPS/JSON for secure data transmission.

For the on-site hardware, the IoT sensor node (ESP32 equipped with environmental sensors) captures ambient metrics and publishes telemetry over MQTT (with TLS) to the Eclipse Mosquitto broker, and the Processing Module subscribes to these topics. The Processing Module performs environmental anomaly detection using either predefined rule-based methods or a pre-trained model accessed from an online/server-based service. No custom model is trained as part of the system. When an anomaly or potential hazard is detected, the Processing Module makes an internal call to the backend server to record the incident and push real-time alert notifications directly to the Admin Dashboard.

Within the Backend, an application server built on Node.js and Express functions as the REST API layer, orchestrating business logic and routing incoming queries. User access control and authentication tokens are managed by an Auth Service utilizing JWT and Supabase Auth. Tabular records such as plant profiles, user accounts, and sensor log are maintained within a PostgreSQL relational database, while high-resolution plant imagery captured in the field is stored in Supabase Object Storage.

4.2.3 Activity Diagram

The activity diagram models the operational workflows of the Smart Ground-Truthing and Digital Biodiversity System, coordinating interactions between five swimlanes: Botanist, System, Visitor, Admin, and Conservation Officer.

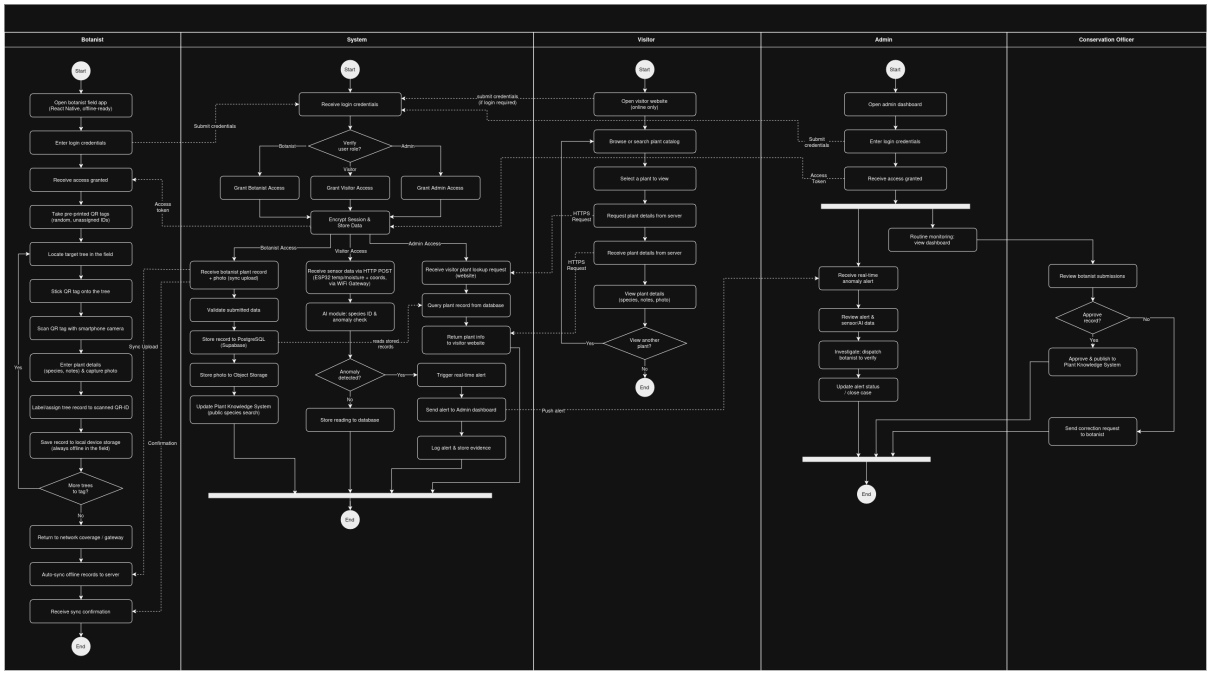


Figure 3: Activity Diagram

Activity_Diagram.drawio

The workflow begins across user roles by opening their respective interfaces such as the React Native Botanist field app, the online Visitor website, the Conservation Officer portal, or the System Administrator dashboard. Botanists, Conservation Officers, and Administrators submit credentials to the central System, while Visitors browse without logging in. The System receives and verifies the user role (Botanist, Conservation Officer, Administrator, or Visitor), grants appropriate access, encrypts the session, and stores data. In the field, a botanist takes pre-printed QR tags with random unassigned IDs, locates target trees, attaches a tag, scans it using a smartphone camera, and enters plant details (species, notes) alongside a captured photo. The specimen record is assigned to the scanned QR-ID and saved to local device storage for offline capability. When network coverage or a gateway is reached, offline records auto-synchronize to the server; the System validates the submitted data, stores records in PostgreSQL, uploads photos to Object Storage, updates the Plant Knowledge System, and sends a sync confirmation back to the mobile app.

For visitors accessing the platform online, plant lookup requests are submitted via HTTPS to the System, which queries the database and returns plant information, notes, and photos for browsing. Concurrently, deployed ESP32 sensor nodes transmit real-time telemetry (temperature, soil moisture, coordinates) to the System over MQTT through the Wi-Fi gateway and Mosquitto broker. The System's anomaly detection module performs anomaly checks; if an anomaly is detected, real-time alerts are triggered and pushed directly to the Administrator

dashboard while logging evidence in the database. Conservation Officers review submitted botanist records and either return them to the botanist with correction requests or approve them for immediate publication in the public Plant Knowledge System. System Administrators monitor the dashboard on a routine basis, and upon receiving real-time anomaly alerts, review the sensor data, dispatch botanists to investigate flagged environmental events, and update case statuses.

4.2.4 Use Cases Diagram

The use case diagram illustrates how different user roles and system components interact with the functional capabilities of the Smart Ground-Truthing and Digital Biodiversity System. The platform defines three human actors (Botanist, Visitor, and Conservation Officer) and two system actors (System Administration and IoT Sensor Node).

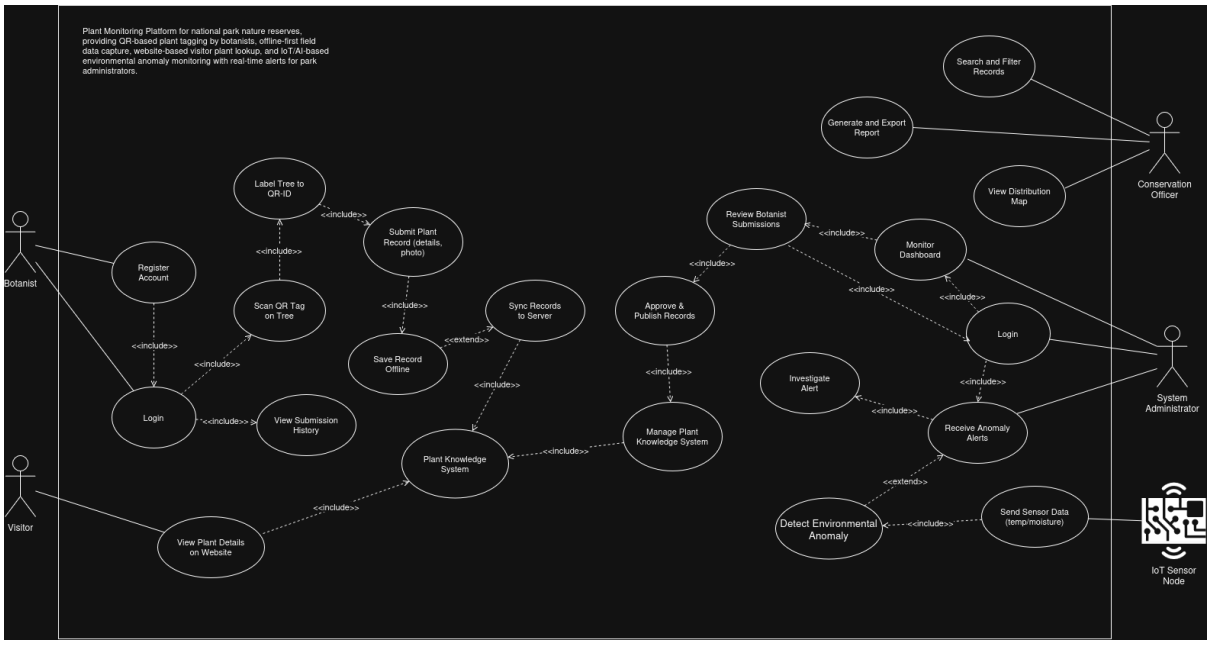


Figure 4: Use Case Diagram

Use_Case_Diagram.drawio

As depicted in the diagram, each actor interacts with specific functionalities based on their access privileges and operational requirements:

- **Botanist:** Registers an account and logs into the field app to manage botanical field surveys. Botanists label trees with QR-IDs, scan QR tags on trees, submit plant records with details and photos, save records offline, sync records to the server and Plant Knowledge System, and view their submission history.
- **Visitor:** Accesses the platform to view plant details on the website via the Plant Knowledge System without requiring an account.
- **Conservation Officer:** Logs into the system to search and filter plant records, generate and export reports, view the distribution map, review botanist submissions, approve

and publish records to the Plant Knowledge System, monitor the dashboard, and manage the Plant Knowledge System.

- **System Administration:** Logs in to monitor the dashboard, receive anomaly alerts, and investigate alerts.
- **IoT Sensor Node:** Sends sensor data (temperature, soil moisture and motion (PIR)) to the system to detect environmental anomalies and trigger alerts.

4.3 Alternative Solutions

Problem Area	Solutions	Evaluation & Justification
Plant ID	• QR Code plaques • RFID/NFC tags • GPS/Location only	RFID/NFC tags cost more, must be encoded with specialised equipment, and NFC has a typical read range of only about 2 cm, so the reader must almost touch the tag (NFC Forum, n.d.). Location only solutions suffer from accuracy problems: smartphone GNSS errors under forest canopy have been measured at roughly 5 to 11 m RMSE (Tomaštík et al., 2017), making the system unable to pinpoint individual plants. The solution chosen is QR Code plaques. QR Codes can be printed easily. They can also be read by most smartphones or have an external camera capture them.
Data Collection & Storage	• Local storage via SQLite and Sync to central PostgreSQL database • IndexedDB • Existing Survey Form collection software (ODK)	IndexedDB is browser-managed storage that can be evicted when the device runs low on space, risking data loss (MDN Web Docs, n.d.). Existing survey form tools such as ODK lack the customisability required for custom dashboards (Hartung et al., 2010). The solution chosen is a local SQLite database stored in the mobile device. This protects data retention in a jungle environment with unreliable network connection. Upon reconnecting to a network, two way conflict resolution can be performed with

		the central PostgreSQL backend server.
Dashboard Interface	• Centralised Web Platform & RESTful API Backend • Spreadsheets & Cloud Database	Spreadsheets alone will scale poorly as this database will also contain photos. The solution chosen is a centralised web platform with a RESTful API backend. This decouples the UI from the data.
Authentication & Access Control	• Role based access control (OAuth 2.0 & JWT) • Server-Side Session Cookies • Single Shared API Key Architecture	Server-side session cookies are inconvenient when people have to operate offline for extended periods in the jungle. Single shared API keys will allow unauthorised individuals full access to the whole database if the key is leaked. The solution chosen is role-based access control via OAuth 2.0 and JSON Web Tokens (Hardt, 2012; Jones et al., 2015). This is a solution that works with offline mobile caching. It also restricts data access based on roles.
IoT Connectivity	• ESP32 microcontroller & MQTT protocol • LoRa • Cellular Data (4G)	LoRaWAN solutions require dedicated gateways and a network server to be installed, adding cost (LoRa Alliance, n.d.). Cellular Data is power hungry and potentially unusable in areas without 4G coverage. The solution chosen is an ESP32 microcontroller running the MQTT protocol to have a low cost, low power solution for real-time sensor data (OASIS, 2019).

Table 9: Alternative Solutions

5 QUALITY MANAGEMENT

Delivering every planned feature is not enough on its own. A feature that loses a botanist's observations when the phone restarts, or reveals where an endangered plant grows, fails the project even if it technically works. For this project, quality means the system is fit for field documentation in Niah National Park: it must not lose data collected offline, field and conservation staff must be able to use it with limited time and connectivity, it must protect sensitive species and user data, and the records it shows must be trustworthy.

To make quality measurable, we use two mechanisms together. The Definition of Done (DoD) describes the state a backlog item must reach before it counts as complete and can go into a sprint increment (Schwaber & Sutherland, 2020). Quality attributes describe how well the item must perform beyond simply working. We take these from the ISO/IEC 25010:2023 product quality model, which defines nine quality characteristics, each with sub-characteristics, for setting acceptance criteria and test objectives (International Organization for Standardization [ISO], 2023). Each attribute has a SMART measurement: specific to one backlog item, measurable by a stated test, achievable for a student prototype, relevant to the brief, and tied to the sprint or week the item is due. An item is only released once it meets both its DoD and its SMART targets.

5.1 Priority Quality Characteristics

The nine ISO/IEC 25010 characteristics do not matter equally for this system. Based on the field conditions and stakeholder needs described earlier, we have prioritised five:

- **Reliability (fault tolerance and recoverability):** Botanists will work offline for long stretches in a forest of around 3,140 hectares with poor connectivity (Meekiong & Wasli, 2025). A record lost to an app crash or a failed sync may be fieldwork that cannot be repeated, so no observation may be lost or duplicated during offline storage or sync.
- **Security (confidentiality, integrity, authenticity and resistance):** The system stores user accounts and the exact locations of protected species, which could help poachers if exposed (Lindenmayer & Scheele, 2017). Following the SSDLC required by the brief, security is assessed, fixed and reassessed throughout, not checked once at the end.
- **Functional suitability (completeness and correctness):** Every feature in the brief's areas of focus must be present and work correctly, because SFC and NeuonAI will assess the prototype against those features.
- **Interaction capability (learnability and operability):** ISO/IEC 25010:2023 uses this term for what the previous edition called usability (ISO, 2023). Field officers enter data outdoors under time pressure, and visitors use the website with no training, so both must be able to finish core tasks without help.
- **Performance efficiency (time behaviour):** IoT alerts only help if they reach the dashboard quickly, and sync must finish within the short windows when a connection is available.

Maintainability, especially modularity and testability, is handled through the development process in the next subsection rather than measured item by item. It matters because the system will be handed over to SFC's administrators when the project ends.

5.2 Quality Assurance Process

Quality is built into every sprint rather than checked at the end. As well as meeting its own criteria in Table 10, every backlog item must pass the following team-wide Definition of Done before the Scrum Master marks it complete in Jira:

- Code is written on a feature branch and merged into main only through a GitHub pull request that at least one other team member has reviewed and approved.
- The item's unit and API tests (Jest and Postman) pass, and no test that passed before now fails.
- npm audit shows no unresolved Critical or High severity vulnerabilities in dependencies.
- The item is shown working in the shared test environment at the sprint review, and the Product Owner accepts it against its acceptance criteria.
- The Jira ticket includes the test evidence, and the matching section of the user manual has been drafted.

Defects found in testing are logged in Jira at one of four severity levels. Critical defects cause data loss, a security breach or a blocked core workflow; High defects make a feature unusable; Medium defects have a workaround; Low defects are cosmetic. An item cannot be marked done while it has open Critical or High defects, but Medium and Low defects can carry over to the next sprint if the Product Owner agrees. Vulnerabilities found in the OWASP ZAP assessment use the same scale and go into a separate remediation log that links each finding to the commit that fixes it. This log is the reassessment evidence the brief asks for.

At the end of Sprint 2 we will run task-based usability tests with participants representing botanists, conservation officers and visitors, followed by the System Usability Scale (SUS). The SUS is a ten-item questionnaire that gives a single score from 0 to 100 (Brooke, 1996), and 68 is commonly used as the average benchmark (Lewis & Sauro, 2018). The unit tutor also reviews each milestone deliverable, and feedback from SFC and NeuonAI goes into the product backlog to be prioritised at the next sprint planning meeting.

5.3 Quality Attributes, SMART Measurement and Definition of Done

Table 10 applies this framework to each item in the product backlog. Item numbers and sprints match the initial release schedule in Section 4. The targets are proposed thresholds for the prototype under controlled test conditions and will be confirmed with the Product Owner at Sprint 1 planning.

Backlog Item (Sprint)	**Quality Attribute (ISO/IEC 25010)**	**Measurement (SMART)**	**Definition of Done**
Project foundation, requirements baseline and shared architecture (Sprint 1)	Maintainability: Modularity Functional suitability: Completeness	By the end of Week 4, the database schema covers 100% of the data fields required by backlog items 2 to 11, and the API contract documents 100% of planned endpoints, verified by a traceability check against the backlog.	Entity relationship diagram and schema reviewed by at least two members and merged; API contract published in the repository; branching strategy, coding conventions and review rules agreed; every member can run the frontend, backend and database locally from the README.
Authentication and role-based access control (Sprint 1)	Security: Authenticity, Confidentiality	By the end of Sprint 1, valid credentials authenticate in 10 of 10 attempts for each role; invalid, expired or tampered tokens are rejected in 10 of 10 attempts; and 10 of 10 attempts to call an endpoint outside the user's role are blocked.	Botanist, Conservation Officer and System Administrator accounts log in on their respective interfaces, while visitor pages need no account; JWTs issued on login and invalidated on logout; role checks enforced at the API, not only in the interface; cached offline sessions revalidated before synchronisation; HTTPS on all endpoints; passwords hashed and personal data encrypted at rest.
Mobile plant observation with local SQLite storage (Sprint 1)	Reliability: Recoverability Functional suitability: Completeness	By the end of Sprint 1, 100% of records created offline across 10 controlled test cases remain intact after the app is force-closed and the device is rebooted, with no field values lost or altered.	Form captures taxonomy, morphological characteristics, height and notes as required by the brief; validation prevents incomplete submission; records and drafts persist in local SQLite; submission history shows each

			record's status (draft, queued, synced or returned).
QR scanning, photographs and GPS capture (Sprint 1)	Functional suitability: Correctness Interaction capability: Operability	By the end of Sprint 1, printed QR tags are read in at least 9 of 10 attempts in both outdoor daylight and shade, and 100% of saved records store GPS coordinates together with their reported accuracy.	Scanning an unassigned tag binds it to a new record and scanning an assigned tag opens the existing record; a QR code can be generated for records without a physical tag; photographs are compressed and stored locally with the record; coordinates are captured automatically.
Cloud synchronisation and conflict/error handling (Sprint 2)	Reliability: Fault tolerance Security: Integrity	By the end of Sprint 2, across 10 test cases where the connection is cut mid-sync, at least 95% of queued records synchronise on the first automatic retry and 100% synchronise eventually, with zero duplicate records.	Sync queue persists across restarts; sync starts automatically when connectivity returns and can be triggered manually; unique record identifiers prevent duplicates; conflicting edits are detected and resolved with an audit trail; interrupted photograph uploads resume; the user sees a clear confirmation or error message.
Web plant database, search and filtering (Sprint 2)	Functional suitability: Completeness Interaction capability: Learnability	By the end of Sprint 2, searches by scientific name, common name and conservation status return the correct records in 20 of 20 test queries, and at least 80% of first-time test participants find and open a named species profile within 3 minutes without help.	Conservation Officers can add, edit and delete species records with taxonomy, description, photographs, conservation status and cultural significance; visitors can search and view published profiles without an account; scanning a public QR tag opens the published profile; no broken links or missing images.
Observation review, mapping and reporting (Sprint 2)	Functional suitability: Correctness Security: Confidentiality	By the end of Sprint 2, approve, reject and return actions produce the correct record state in 10 of	Review queue lists pending botanist submissions; correction requests reach the botanist's app; only

		10 test cases; 100% of endangered species records show reduced-precision coordinates in public views; and 5 of 5 test exports open without error and match the on-screen data.	approved records are published to the knowledge system; the distribution map renders from approved coordinates; reports export to CSV and PDF.
ESP32 IoT sensing, MQTT ingestion and data storage (Sprint 2)	Reliability: Availability Performance efficiency: Time behaviour	By the end of Sprint 2, the sensor node publishes continuously for at least 60 minutes without disconnection in 3 consecutive runs, and each reading is stored in the database within 10 seconds of publication.	Each fitted sensor (temperature, humidity, soil moisture, movement and GPS) returns plausible readings in bench testing; firmware publishes over MQTT with TLS to the Mosquitto broker; node disconnection is detected; the backend stores readings with node ID and timestamp; a telemetry simulator is available when hardware is not.
IoT dashboard, threat detection and automated alerts (Sprint 2)	Performance efficiency: Time behaviour Functional suitability: Correctness	By the end of Sprint 2, the median time from a controlled sensor event to a dashboard alert is under 10 seconds on a stable network; at least 8 of 10 simulated disturbance events raise an alert; and no more than 2 false alerts occur in a 24-hour baseline run using recorded or simulated telemetry.	Dashboard shows live telemetry per node; alert thresholds are configurable per node; alerts require a sustained breach rather than a single spike; each alert is logged with timestamp, node and reading; acknowledgement records who responded and when; case status can be updated to resolved.
Security assessment, usability testing and integrated demonstration (Sprint 2)	Security: Resistance Interaction capability: Operability Functional suitability: Completeness	By Week 11, zero Critical or High findings remain unresolved after reassessment, with 100% of findings either fixed or given a documented risk acceptance; at least	Vulnerability assessment run with OWASP ZAP and npm audit against a documented scope; findings rated by severity; remediation log links each finding to its fix; reassessment report shows before-and-after

		95% of planned functional and integration test cases pass with no open Critical defects; and a mean SUS score of at least 68 is achieved from at least 5 participants.	results; the end-to-end flow (tag, record offline, sync, review, publish, visitor view) is demonstrated; defect log and user manual complete.
Optional ground-truthing innovation and observation quality indicator (Sprint 2)	Functional suitability: Appropriateness	By Week 11, the quality indicator scores 100% of submitted records, and a timed comparison against the manual paper workflow on at least 10 sample records is documented, reporting recording time and the number of missing fields for each method.	Indicator is shown to reviewers alongside each submission; scoring rules and evaluation method are documented; results and limitations are reported. This item may be descoped if any mandatory item is incomplete entering the final two weeks.

Table 10: Quality attributes, SMART measurement and Definition of Done for each backlog item

An item that misses a target at the sprint review is not counted as done. It goes back into the backlog with the failed result recorded, and we decide with the Product Owner whether to fix it in the next sprint, revise the target with a written justification, or drop it.

6 RESOURCES

This section outlines the resources needed to complete this project. The resources are divided into human resources, software resources and hardware resources, and a cost analysis of the working prototype. Human resources describe the knowledge, skills and attributes of the project team. Software and hardware resources list the tools, platforms and components used to build and test the system. The resources listed are based on the current project scope and plan. They are subject to change as the project evolves and development progresses.

6.1 Human Resources

Human resources refer to the knowledge, skills and attributes that the seven project team members bring to the development of the system.

6.1.1 Knowledge

- Basic understanding of IoT systems and architectures
- Understanding of offline data storage and cloud synchronisation
- Knowledge of RESTful API design and secure software development (SSDLC)

6.1.2 Abilities and Skills

- Ability to develop web and mobile applications using React and React Native
- Ability to develop backend systems using Node.js (Express) and Supabase
- Ability to integrate IoT hardware such as ESP32 and sensors using MQTT
- Knowledge of database management (PostgreSQL and SQLite)
- Ability to debug, test, and secure the system using Postman, Jest, and OWASP ZAP

6.1.3 Attitudes

- Strong teamwork and collaboration
- Willingness to learn and adapt to new technologies
- Responsible and proactive in completing assigned tasks

6.1.4 Energy

- Ability to dedicate at least 4-6 hours per week to project development

6.1.5 Time

- Three weeks for requirements analysis, solution analysis and proposal preparation.
- Approximately eight weeks for development across two sprints, followed by integration, testing and documentation.

6.2 Software Resources

	Resource Name	What Resource is For
1	Visual Studio Code	Integrated development environment for all application development
2	React with Vite and React Router	Web frontend for the knowledge system, administrator dashboard and public visitor view
3	React Native with Expo	Cross-platform mobile framework for the offline-capable Botanist Field Application
4	Axios	HTTP client for HTTPS and JSON communication between clients and the API
5	SQLite (expo-sqlite)	On-device storage for offline plant records, drafts and the synchronisation queue
6	Node.js with Express	Backend REST API layer and business logic
7	Supabase Auth	Authentication, JSON Web Token issuance and session lifecycle management
8	PostgreSQL via Supabase	Central relational database for taxonomy, specimens, observations, users, nodes, telemetry and alerts
9	Supabase Object Storage	Storage for botanical photographs and motion-triggered sensor images
10	Arduino IDE or PlatformIO	ESP32 firmware development and flashing
11	Eclipse Mosquitto	MQTT broker for sensor telemetry ingestion over TLS
12	MQTT Explorer	Inspection and debugging of broker topics during development
13	QR encoding and scanning libraries	QR code generation for new records and camera-based scanning
14	Python with scikit-learn	Statistical anomaly detection over sensor time series

15	Leaflet with OpenStreetMap	Species distribution map rendering
16	Postman and Jest	API testing, unit testing and integration testing
17	OWASP ZAP and npm audit	Dynamic application security testing and dependency vulnerability scanning
18	Git and GitHub	Version control, pull request review and collaborative development
19	Jira	Sprint backlog tracking, task assignment and timeline management
20	draw.io	Interface wireframing and system architecture diagramming

Table 11: Software Resources

6.3 Hardware Resources

	Resource Name	**What Resource is For**
1.	Team laptops and desktop computers	Development, testing and documentation
2.	Android smartphones	Testing the Botanist Field Application and visitor access, including camera, GPS, offline storage and synchronisation behaviour
3.	ESP32-CAM module	Sensor node controller providing processing, wireless connectivity and motion-triggered image capture
4.	DHT22 temperature and humidity sensor	Ambient temperature and humidity readings required by the project brief
5.	GPS module (NEO-6M)	Geolocation of the sensor node and its associated specimen
6.	Capacitive soil moisture sensor	Habitat condition readings supporting long-term ecological analysis
7.	Breadboard, jumper wires and resistors	Prototype assembly and sensor interfacing

8.	Portable power bank	Wire-free power for field demonstration of the sensor node
9.	Wireless router or mobile hotspot	Connectivity gateway between the sensor node and the backend
10.	Pre-printed waterproof QR tags	Physical labelling of selected specimens during field testing
11.	PIR motion sensor (HC-SR501)	Detects movement near the monitored plant and triggers ESP32-CAM image capture.

Table 12: Hardware Resources

6.4 Cost Analysis of Prototypes

Based on the current project scope, the information below is the estimated cost for a working prototype. The prices are for a single prototype only and are not the estimated price of deployment.

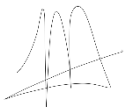
	Components	**Quantity**	**Estimated Price (RM)**	**Remark**
1.	ESP32-CAM module	1	40	May be available from the IoT laboratory
2.	DHT22 temperature and humidity sensor	1	15	Replaces a separate temperature-only sensor
3.	GPS module (NEO-6M)	1	35	Required for node and specimen geolocation
4.	Capacitive soil moisture sensor	1	12	Supports long-term habitat condition analysis
5.	Breadboard, jumper wires and resistors	1 set	15	Prototype assembly
6.	Portable power bank	1	40	Free if a team member already owns one
7.	Waterproof QR tag printing	20	15	Printed on weather-resistant sticker stock

8.	Spare sensors for fault contingency	1 set	25	Contingency for sensor faults
9.	Supabase (database, auth, storage)	1	Free	Free tier sufficient for prototype data volumes
10.	Mosquitto broker hosting	1	Free	Runs locally during development and demonstration
11.	PIR motion sensor (HC-SR501)	1	5	Detects movement to trigger alerts and image capture
12.	All other software	–	Free	Open source or free tier as listed in Table 5
	Estimated total		202	One-off, with no recurring cost

Table 13: Cost Analysis of Prototypes

7 APPROVAL SIGNATURES

Project Team

	Name of student	Student Id	Signature	Roles
1	Amberlyn Xi En LIM	104391342		Developer
2	Albian Pang SEO	104392785		Product Owner, Developer
3	Michelle Mei Xuan TAN	104394341		Developer
4	Nicole Jia Xin TAN	104406767		Scrum Master, Developer
5	Norman Zhi Wen CHUNG	104387930		Developer
6	Patrick Lip Siang SIM	104393720		Developer
7	Ricardo Zhen Wei CHEW	104392756		Developer

Project Sponsor [Ts. Dr. Lee Sue Han]

Tutor's name (on behalf of the client)	Signature:
Ts. Dr. Lee Sue Han	

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